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A note on the power prior

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SUMMARY

The power prior by Ibrahim and Chen (*Statist. Sci.* 2000; **15**:46–60) is one of several methods to incorporate historical data in the analysis of a clinical trial. The power prior raises the likelihood of the historical data to the power parameter a_0 which quantifies the discounting of the historical data due to heterogeneity between trials. It is shown that the standard method of estimating the power parameter from the historical and current data is inappropriate, and we therefore suggest to use a modified power prior approach or to consider alternative methods instead. Copyright © 2009 John Wiley & Sons, Ltd.

KEY WORDS: clinical trial; discounting; historical data; heterogeneity; meta-analysis

1. INTRODUCTION

The quantification of information arising from historical data is an important topic, for example, in clinical trials where historical control data are sometimes incorporated into the analysis of a trial, particularly in the early phases of drug development. The *power prior* has been proposed in this context [1–3] as a flexible approach that allows the user to introduce an appropriate weight that discounts historical data. While the power approach is an important tool to discount historical data, we will show that one of the proposals for estimating the weight (the power parameter) is unwarranted and may lead to inappropriate inferences.

Following the standard setup and notation of the power prior approach, assume that for a parameter θ there will be data D from the current trial, and also historical data D_0. The power prior introduces historical information D_0 by the likelihood of D_0 raised to a power a_0, that is,

$$\pi(\theta|D_0, a_0) \propto L(\theta|D_0)^{a_0} \pi_0(\theta) \tag{1}$$

where the normalizing constant, that depends on D_0 and a_0, is omitted. The initial prior $\pi_0(\theta)$ is often chosen to be noninformative. The special cases of using the historical data fully or not at

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all are covered by $a_0=1$ and $a_0=0$, respectively, while values of a_0 between 0 and 1 allow for differential weighting of the historical data. For example, in the case of n normally distributed observations and with mean θ, it is easy to see that $L(\theta|D_0)^{a_0}$ is equivalent to a likelihood arising from the same sufficient statistic (mean of observations) but with a variance that is inflated by $1/a_0$, or, equivalently, a discounted historical sample size of $n_0=a_0n$. Similar interpretations apply for other members of the exponential family, for example, the binomial, Poisson, and exponential distribution.

The formulation (1) assumes a_0 as known. For example, elicitation of specific values of a_0 can be done by prior elicitation of expert opinions, or via a meta-analytic argument that assumes the historical and current parameter as exchangeable. For the relationship between the power prior parameter a_0 and the random-effects between-trial standard deviation τ, see Chen and Ibrahim [4].

The extension of (1) to the case where uncertainty about a_0 is expressed by a prior distribution requires a joint prior $\pi(\theta, a_0)$ for the two parameters θ and a_0. The following prior has been suggested [1–3]:

$$\pi(\theta, a_0|D_0) \propto L(\theta|D_0)^{a_0}\pi_0(\theta)\pi(a_0) \tag{2}$$

We suggest that this prior is inappropriate and has undesirable properties. In (2) the joint prior for (θ, a_0) is decomposed into the marginal prior $\pi(a_0)$ of a_0, for example, a uniform prior between 0 and 1, and the power prior as in (1), that is, the conditional prior of θ given the historical data and a_0 as in (1). As in (2) a_0 is now an additional parameter, the normalizing constant in (1) that depends on a_0 cannot be ignored in (2). Essentially, (2) misses a factor that depends on a_0, namely

$$C(a_0)=1 \Big/ \int L(\theta|D_0)^{a_0}\pi_0(\theta)\,\mathrm{d}\theta \tag{3}$$

Therefore, the power prior for the case where a_0 is treated as an unknown parameter should be of the form

$$\pi(\theta, a_0|D_0) = C(a_0)L(\theta|D_0)^{a_0}\pi_0(\theta)\pi(a_0) \tag{4}$$

The difference between (2) and (4) is important. In (4) the marginal prior of θ can be obtained by averaging the conditional prior of θ given a_0 as in (1), that is, $L(\theta|D_0)^{a_0}\pi_0\,(\theta)$, over the marginal prior of a_0. This is not possible under (2), as the weights $C(a_0)$ have been omitted. Standard probability calculus, which is necessary in the Bayesian framework, is therefore only applicable using (4).

A simple example that allows for an analytic posterior of a_0 shows the difference in results when using (2) instead of (4). Assume binary data with sample sizes n_0 and n for the historical and current trial. Let x_0 andy_0 be the number of historical responders and nonresponders, and assume that the respective numbers for the current study are x and y.

If we assume a Beta(c,d) prior for θ, from (2) it then follows that the joint prior $\pi(\theta, a_0|x_0, y_0)$ is proportional to

$$\theta^{a_0x_0+c-1}(1-\theta)^{a_0y_0+d-1}\pi(a_0) \tag{5}$$

However, using (4) with the normalizing constant in (1) leads to the joint prior of the form

$$\pi(\theta, a_0|x_0, y_0) = \frac{\Gamma(a_0n_0+c+d)}{\Gamma(a_0x_0+c)\Gamma(a_0y_0+d)}\theta^{a_0x_0+c-1}(1-\theta)^{a_0y_0+d-1}\pi(a_0) \tag{6}$$

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Table I. Posterior mean (95 per cent interval) for a_0 for four data scenarios of historical (H) and current (C) data; uniform priors for response rate θ and power parameter a_0.

	Historical (H) and current (C) Data x/n		Posterior mean (95 per cent interval) from (2), (7)	Posterior mean (95 per cent interval) from (4), (8)
Scenario 1	H	$\frac{20}{100}$	0.02 (0.00,0.07)	0.57 (0.07,0.98)
	C	$\frac{20}{100}$		
Scenario 2	H	$\frac{10}{100}$	0.03 (0.00,0.10)	0.43 (0.03,0.96)
	C	$\frac{200}{1000}$		
Scenario 3	H	$\frac{200}{1000}$	0.00 (0.00, 0.01)	0.58 (0.07,0.98)
	C	$\frac{200}{1000}$		
Scenario 4	H	$\frac{100}{1000}$	0.00 (0.00, 0.01)	0.05 (0.00,0.15)
	C	$\frac{200}{1000}$		

From (2), the marginal posterior distribution for a_0 is obtained as

$$f(a_0|x_0, y_0, x, y) = \frac{\Gamma(a_0x_0+x+c)\Gamma(a_0y_0+y+d)}{\Gamma(a_0n_0+n+c+d)}\pi(a_0) \tag{7}$$

On the other hand, if (4) is used, the marginal posterior distribution is

$$f(a_0|x_0, y_0, x, y) = \frac{\Gamma(a_0n_0+c+d)\Gamma(a_0x_0+x+c)\Gamma(a_0y_0+y+d)}{\Gamma(a_0x_0+c)\Gamma(a_0y_0+d)\Gamma(a_0n_0+n+c+d)}\pi(a_0) \tag{8}$$

Four data scenarios are presented in Table I. For the historical and current trial, the scenarios assume equal response rates (20 per cent), or different response rates (20 and 10 per cent). The sample sizes for the two trials are assumed equal (100 or 1000). We would expect that the data that show evidence of similar historical and current response rates would lead to a larger a_0 and hence increased pooling of the data, whereas that with evidence of discordant historical and current response rates would lead to lower a_0 and hence decreased use of the historical data.

We consider the correct version (4) first. The results show that if the two trials are small ($n_0 = n = 100$), the posterior of a_0 is close to the prior regardless of the choice of data. This is not surprising as only two data sets are available and the heterogeneity of the data cannot be assessed precisely. A similar explanation can be obtained using the relationship between the power prior and a hierarchical (random-effects meta-analytic) model: in a random-effects meta-analysis it is difficult to obtain a reasonably precise estimate for the between-trial variability, in particular if only a few small trials are available. The results are somewhat more encouraging if both trials are large ($n_0 = n = 1000$) and the two response rates differ (scenario 4), the posterior is markedly shifted downwards compared with the uniform prior, with a posterior interval for a_0 ranging from 0 to 0.15.

If version (2) of the power prior is used, the results are very different and highly unintuitive. The posterior is now very precise and concentrated close to 0 leading to almost complete rejection of

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the historical data even if the historical and current response rates are the same. This is in contrast to the fact that data from two trials cannot be very informative as to the heterogeneity of the data.

In this simple example the consequences arising from the omission of the normalizing constant in (2) are obvious. However, things may not be so clear when more complicated models are used. We use Example 1 in Ibrahim and Chen [1] to illustrate the argument, but other examples in the literature could be used as well. In their example the sample sizes of the historical and actual study were $n_0=823$ and $n=183$ and the data were analyzed by logistic regression. Posterior means for a_0 under different prior assumptions were presented (Table I in Ibrahim and Chen [1]). One of the extreme priors used to assess sensitivity of results assumed a Beta$(100, 1)$ prior, with a corresponding posterior mean a_0 of 0.53. This is implausible, as *a priori* the 95 per cent interval for a_0 is $(0.96, 1.0)$ and it would be very surprising if the data from the two trials could shift this evidence substantially. The only explanation is the use of (2) instead of (4). The substantial shifting of the posterior downwards is similar to the example in Table I.

We would like to note that although most applications we found in the literature seem to make use of the prior in (2), there are two notable exceptions [5, 6]. In fact the 'modified power prior approach' discussed in Duan (unpublished dissertation) introduces the missing normalizing constant in (4). However, it seems not to have been recognized in the statistical literature as the appropriate implementation of the power prior approach when a_0 is unknown.

In summary, the use of the power prior is a valuable tool to incorporate and discount historical data. The power parameter a_0 is easy to interpret, but its elicitation is challenging. We would not particularly recommend taking a_0 as an unknown parameter, but if this is desired then the 'original' prior for a_0 as originally suggested in (2) appears inappropriate, and the use of the 'modified' prior (4) is needed in order to be compliant with standard Bayesian probabilistic arguments.

The power approach depends on the availability of a prior for a_0 and the computability of the normalizing constant (3). As a default the former might be obtained via a random-effects meta-analytic approach [4], whereas the latter may be available in closed form in some circumstances or may require numerical approximation or Markov chain Monte Carlo (MCMC) methods. In such a setting, power priors can be thought of as a means to center prior distributions on historical point estimates, with the same functional form as the likelihood but with an expanded variance. The resulting posterior will be based essentially on a weighted average of the historical and current evidence, regardless of any conflict between the historical and current data. An alternative would be to form a 'robust' prior centered on the historical data but with, say, a Student t form. David [7] and O' Hagan [8] showed this would ensure that in the event of substantial conflict between historical and current data, the historical data would be essentially ignored.

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